

IF440: Artificial Intelligence

Topic 9: Expectation Maximisation Algorithm

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M U L T I M E D I A
N U S A N T A R A

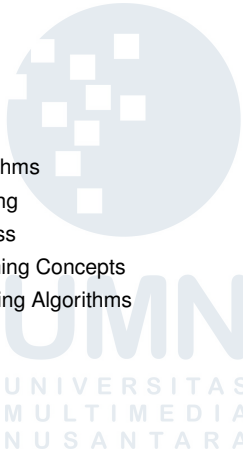
Course Sub Learning Outcomes (Sub-CLO):



SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).



Review

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- 1 Machine Learning Algorithms
 - 2 Types of Machine Learning
 - 3 Machine Learning Process
 - 4 Important Machine Learning Concepts
 - 5 Common Machine Learning Algorithms

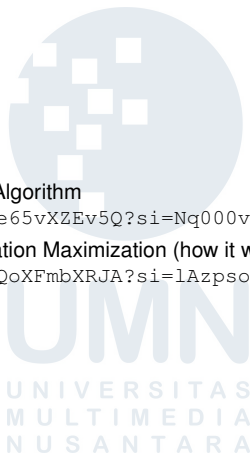
Pre-class Assignment

- Francisco Iacobelli: EM Algorithm

<https://youtu.be/7e65vXZEv5Q?si=Nq000vGqAQd-duJH>

- Victor Lavrenko: Expectation Maximization (how it works)

<https://youtu.be/iQoXFmbXRJA?si=lAzpso0PdYXaJZ51>



Outline

- 
- 1 Introduction to Expectation-Maximisation algorithm
 - 2 Steps of Expectation-Maximisation algorithm
 - 3 Example of Expectation-Maximisation algorithm
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Introduction to Expectation-Maximisation algorithm



INTRODUCTION

- An iterative algorithm for maximizing likelihood estimation (MLE) when the model contains unobserved latent variables
- Was initially invented by computer scientist in special circumstances
- Generalized by Arthur Dempster, Nan Laird, and Donald Rubin in a classic 1977 JRSSB paper, which is widely known as the “DLR” paper

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Maximum Likelihood from Incomplete Data via the EM Algorithm

By A. P. DEMPSTER, N. M. LAIRD and D. B. RUBIN

Harvard University and Educational Testing Service

[Read before the ROYAL STATISTICAL SOCIETY at a meeting organized by the RESEARCH SECTION on Wednesday, December 8th, 1976, Professor S. D. SILVEY in the Chair]

SUMMARY

A broadly applicable algorithm for computing maximum likelihood estimates from incomplete data is presented at various levels of generality. Theory showing the monotone behaviour of the likelihood and convergence of the algorithm is derived. Many examples are sketched, including missing value situations, applications to grouped, censored or truncated data, finite mixture models, variance component estimation, hyperparameter estimation, iteratively reweighted least squares and factor analysis.

Keywords: MAXIMUM LIKELIHOOD; INCOMPLETE DATA; EM ALGORITHM; POSTERIOR MODE

1. INTRODUCTION

THIS paper presents a general approach to iterative computation of maximum-likelihood estimates when the observations can be viewed as incomplete data. Since each iteration of the algorithm consists of an expectation step followed by a maximization step we call it the EM algorithm. The EM process is remarkable in part because of the simplicity and generality of the associated theory, and in part because of the wide range of examples which fall under its umbrella. When the underlying complete data come from an exponential family whose maximum-likelihood estimates are easily computed, then each maximization step of an EM algorithm is likewise easily computed.

The term "incomplete data" in its general form implies the existence of two sample spaces $\mathcal{Y}$ and $\mathcal{X}$ and a many-one mapping from $\mathcal{X}$ to $\mathcal{Y}$. The observed data y are a realization from $\mathcal{Y}$. The corresponding x in $\mathcal{X}$ is not observed directly, but only indirectly through y . More specifically, we assume there is a mapping $x \rightarrow y(x)$ from $\mathcal{X}$ to $\mathcal{Y}$, and that x is known only to lie in $\mathcal{X}(y)$, the subset of $\mathcal{X}$ determined by the equation $y = y(x)$, where y is the observed data. We refer to x as the *complete data* even though in certain examples x includes what are traditionally called parameters.

We postulate a family of sampling densities $f(x|\phi)$ depending on parameters ϕ and derive its corresponding family of sampling densities $g(y|\phi)$. The complete-data specification $f(\dots|\dots)$ is related to the incomplete-data specification $g(\dots|\dots)$ by

$$g(y|\phi) = \int_{\mathcal{X}(y)} f(x|\phi) dx. \quad (1.1)$$

The EM algorithm is directed at finding a value of ϕ which maximizes $g(y|\phi)$ given an observed y , but it does so by making essential use of the associated family $f(x|\phi)$. Notice that given the incomplete-data specification $g(y|\phi)$, there are many possible complete-data specifications $f(x|\phi)$ that will generate $g(y|\phi)$. Sometimes a natural choice will be obvious, at other times there may be several different ways of defining the associated $f(x|\phi)$.

Each iteration of the EM algorithm involves two steps which we call the expectation step (E-step) and the maximization step (M-step). The precise definitions of these steps, and their associated heuristic interpretations, are given in Section 2 for successively more general types of models. Here we shall present only a simple numerical example to give the flavour of the method.

MLE

- MLE involves treating the problem as **an optimization or search problem**, where we seek a set of parameters that results in the best fit for the joint probability of the data sample ($\mathcal{X}$).
- It involves maximizing a likelihood function in order to find the probability distribution and parameters that best explain the observed data.



Steps of Expectation-Maximisation algorithm

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INTRODUCTION

- The algorithm iterate between E-Step (expectation) and M-STEP (maximization)
- **E-STEP** : create a function for the expectation of the log-likelihood, evaluated using the current estimate for the parameters
- **M-STEP** : compute parameters maximizing the expected log-likelihood found in the E-STEP

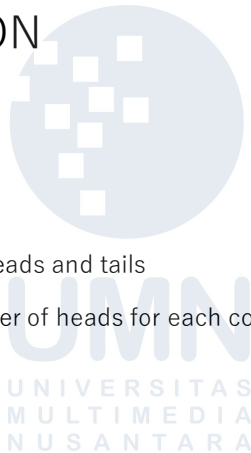
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SOME INTUITION

- Suppose you have 2 coins, A and B, each with a certain bias of landing heads, $\hat{\theta}_A$ and $\hat{\theta}_B$.
- If you pick one at random and toss it. Which one was it?

SOME INTUITION

- Let's do this five times:
 - Pick a coin randomly
 - Toss it 10 times
 - Record the number of heads and tails
- Then get the average number of heads for each coin



Cuong B. Do and Serafim Batzoglou (2008)

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A COIN FLIPPING EXPERIMENT : Simplified MLE

B	H	T	T	T	H	H	T	H	T	H
A	H	H	H	H	T	H	H	H	H	H
A	H	T	H	H	H	H	H	T	H	H
B	H	T	H	T	T	T	H	H	T	T
A	T	H	H	H	T	H	H	H	T	H

5 sets, 10 tosses per set

Sum :

Coin A	Coin B
	5 H, 5 T
9 H, 1 T	
8 H, 2 T	
7 H, 3 T	
24 H, 6 T	9 H, 11 T

$$\hat{\theta}_A = \frac{24}{24 + 6} = 0.80$$

$$\hat{\theta}_B = \frac{9}{9 + 11} = 0.45$$

- $\hat{\theta}$ The probability of getting heads
- $\hat{\theta}_A$ The probability of coin A landing on head
- $\hat{\theta}_B$ The probability of coin B landing on head

A COIN FLIPPING EXPERIMENT : Simplified MLE

B	H	T	T	T	H	H	T	H	T	H
A	H	H	H	H	T	H	H	H	H	H
A	H	T								
B	H	T								
A	T	H								

Coin A	Coin B
9 H, 1 T	5 H, 5 T

$$\hat{\theta}_A = \frac{24}{24 + 6} = 0.80$$

$$\hat{\theta}_B = \frac{9}{9 + 11} = 0.45$$

Goal : determine coin parameters without knowing the identity of each data set's coin.

Solution : Expectation Maximization

$\hat{\theta}$ The probability of getting heads

$\hat{\theta}_A$ The probability of coin A landing on head

$\hat{\theta}_B$ The probability of coin B landing on head

A COIN FLIPPING EXPERIMENT: Question

- That was easy : Coin A yields heads 80% of the time, Coin B 45% of the time
- What if we are given ONLY the result of our coin tosses
- Can we guess the percentage of heads that each coin yields?
- Can we guess which coin was picked for each set of 10 coin tosses?

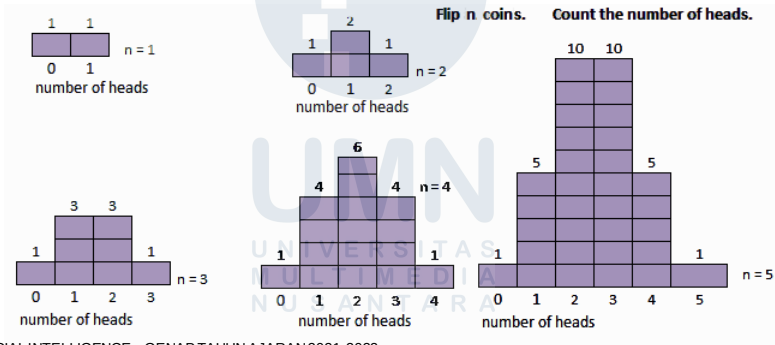
A COIN FLIPPING EXPERIMENT:

One way to think about this is :

1. Assign random averages to both coins
2. For each of the 5 rounds of 10 coin tosses
 - Check the percentage of heads
 - Find the probability of it coming from each coin
 - Compute the expected number of heads : using that probability as a weight, multiply it by the number of heads
 - Record those numbers
 - Re-compute new means for coin A and B
3. With these new means go back to step 2

HOW DO COIN TOSSES BEHAVE?

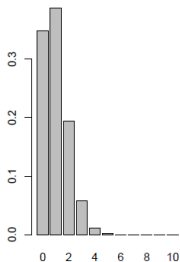
Binomial Distribution



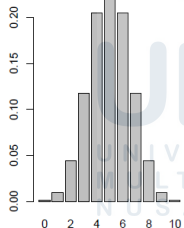
HOW DO COIN TOSSES BEHAVE?

Binomial Distribution

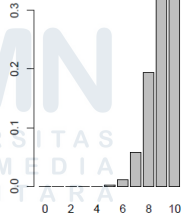
Binomial $n=10$ $p=0.1$



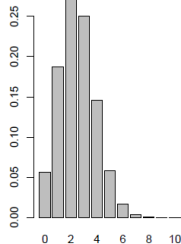
Binomial $n=10$ $p=0.5$



Binomial $n=10$ $p=0.9$



Binomial $n=10$ $p=0.25$





Example of Expectation-Maximisation algorithm

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COIN FLIP WITH HIDDEN VARIABLES

- What if you were given the same dataset of coin flip results, but no coin identities defining the datasets?

Here : $X = \{x_1, \dots, x_m\}$, the observed variable

$$Z = \begin{Bmatrix} z_{1,1} & \dots & z_{m,1} \\ \dots & z_{i,j} & \dots \\ z_{1,k} & \dots & z_{m,k} \end{Bmatrix} \text{ where } z_{i,j} = \begin{cases} 1; & \text{if } x_i \text{ is from } j^{\text{th}} \text{ coin} \\ 0; & \text{otherwise} \end{cases}$$

But Z is not known. (i.e. : 'hidden' / 'latent' variable)

EM ALGORITHM

a) Initialize some arbitrary hypothesis of parameter values ($\hat{\theta}$) :

$$\hat{\theta} = \{\hat{\theta}_1, \dots, \hat{\theta}_k\} \quad \text{coin flip example : } \hat{\theta} = \{\hat{\theta}_A, \hat{\theta}_B\} = \{0.6, 0.5\}$$

b) Expectation (E-STEP)

$$E[z_{i,j}] = \frac{p(x = x_i | \hat{\theta} = \hat{\theta}_j)}{\sum_{n=1}^k p(x = x_i | \hat{\theta} = \hat{\theta}_n)}$$

c) Maximization (M-STEP)

$$\hat{\theta}_j = \frac{\sum_{i=1}^m E[z_{i,j}] x_i}{\sum_{i=1}^m E[z_{i,j}]}$$

EM ALGORITHM

H	T	T	T	H	H	T	H	T	H
H	H	H	H	T	H	H	H	H	H
H	T	H	H	H	H	H	T	H	H
H	T	H	T	T	T	H	H	T	T
T	H	H	H	T	H	H	H	T	H

5 sets, 10 tosses per set

- Initialize $\hat{\theta}_A, \hat{\theta}_B$ to the chosen value (ex : $\hat{\theta}_A = 0.6, \hat{\theta}_B = 0.5$)
- Compute a probability distribution of possible completions of the data using current parameters

EM ALGORITHM

Set 1

H	T	T	T	H	H	T	H	T	H
---	---	---	---	---	---	---	---	---	---

- What is the probability that I observe 5 heads and 5 tails in coin A and B given the initializing parameters $\hat{\theta}_A = 0.6, \hat{\theta}_B = 0.5$?

EM ALGORITHM

Set 1

H	T	T	T	H	H	T	H	T	H
---	---	---	---	---	---	---	---	---	---

- Compute likelihood of set 1 coming from coin A or B using the binomial distribution with mean probability $\hat{\theta}$ on n trials with k successes

$$p(k) = \binom{n}{k} \hat{\theta}^k (1 - \hat{\theta})^{n-k}$$

- Lets take the first round : 5/10 heads and 5/10 tails
- Likelihood of "A" = $P_A(h)^h * (1 - P_A(h))^{10-h} = 0.00079$
- Likelihood of "B" = $P_B(h)^h * (1 - P_B(h))^{10-h} = 0.00097$
- Normalize to get probabilities $\rightarrow A = 0.45, B = 0.55$

EM ALGORITHM

Set 1

H	T	T	T	H	H	T	H	T	H
---	---	---	---	---	---	---	---	---	---

So, we have :

$$\hat{\theta}_A = 0.6, \hat{\theta}_B = 0.5$$

Recap : $P(\text{Coin} = A) = 0.45$; $P(\text{Coin} = B) = 0.55$

E-step - Estimating likely number of heads and tails from :

- “A” : $H = 0.45 \times 5 \text{ heads} = 2.2 \text{ heads}$; $T = 0.45 \times 5 \text{ tails} = 2.2 \text{ tails}$
- “B” : $H = 0.55 \times 5 \text{ heads} = 2.8 \text{ heads}$; $T = 0.55 \times 5 \text{ tails} = 2.8 \text{ tails}$

Do the same for all five runs

THE E-STEP



- Compute the new probabilities for each coin ($\frac{H}{H+T}$)
- That gives you the new maximized parameter θ for each coin

THE M-STEP

Coin A	Coin B
$\approx 2.2 \text{ H}, 2.2 \text{ T}$	$\approx 2.8 \text{ H}, 2.8 \text{ T}$
$\approx 7.2 \text{ H}, 0.8 \text{ T}$	$\approx 1.8 \text{ H}, 0.2 \text{ T}$
$\approx 5.9 \text{ H}, 1.5 \text{ T}$	$\approx 2.1 \text{ H}, 0.5 \text{ T}$
$\approx 1.4 \text{ H}, 2.1 \text{ T}$	$\approx 2.6 \text{ H}, 3.9 \text{ T}$
$\approx 4.5 \text{ H}, 1.9 \text{ T}$	$\approx 2.5 \text{ H}, 1.1 \text{ T}$
$\approx 21.3 \text{ H}, 8.6 \text{ T}$	$\approx 11.7 \text{ H}, 8.4 \text{ T}$

$$\widehat{\theta}_A(1) = \frac{21.3}{21.3 + 8.6} = 0.71$$

$$\widehat{\theta}_B(1) = \frac{11.7}{11.7 + 8.4} = 0.58$$

Repeat E-STEP and M-STEP until convergence

The convergence stop at :

$$\widehat{\theta}_A(10) = 0.80$$

$$\widehat{\theta}_B(10) = 0.52$$

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SUMMARY

- Choose starting parameters
- Estimate probability using these parameter using that each data set (x_i) came from j^{th} coin ($E[z_{i,j}]$)
- Use these probability values ($E[z_{i,j}]$) as weight on each data point when computing a new θ_j to describe each distribution
- Summate these expected values, use maximum likelihood estimation to derive new parameter values to repeat process

Applications of EM Algorithm

- Clustering: Gaussian Mixture Model

<https://builtin.com/articles/gaussian-mixture-model>

- Bioinformatics and Gene Expression Analysis

- Anomaly Detection in Cybersecurity

- Image Segmentation for Medical Imaging

- Sentiment Analysis on Social Media

- Fraud Detection in Financial Transactions

source: <https://www.numberanalytics.com/blog/10-real-world-applications->

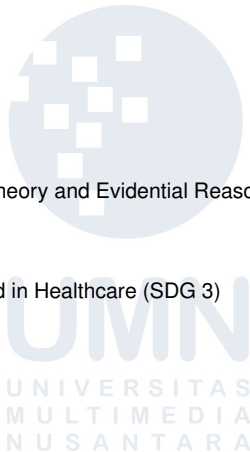
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- [1] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum Likelihood from Incomplete Data Via the EM Algorithm," **Journal of the Royal Statistical Society Series B: Statistical Methodology**, vol. 39, no. 1, pp. 1–22, 9 1977. [Online]. Available: <https://dx.doi.org/10.1111/j.2517-6161.1977.tb01600.x>
- [2] C. B. Do and S. Batzoglou, "What is the expectation maximization algorithm?" **Nature Biotechnology** 2008 26:8, vol. 26, pp. 897–899, 8 2008. [Online]. Available: <https://www.nature.com/articles/nbt1406>

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Next Week

- Bias of Bayesian Method
- Certainty Factors (CF) Theory and Evidential Reasoning
- Example of CFs Theory
- Dempster-Shafer Theory
- Application of DS Method in Healthcare (SDG 3)
- Exercises



Vision and Mission

Vision

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Mission

- 1 Organising learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.

